# Ege Çırakman

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## Education

Istanbul Technical University (ITU), Istanbul, Turkey — *Expected June 2026*
B.Sc., Control & Automation Engineering
GPA: 3.71/4.00 | Department Rank: 1/101 (as of 2024)

TED Ankara College Foundation Private High School, Ankara, Turkey

## Publications

*Peer-Reviewed Publications*

**E. Çırakman***, H. T. Erdinc*, F. J. Herrmann. *Efficient and scalable posterior surrogate for seismic inversion via wavelet score-based generative models*. IMAGE 2025 (Oral Presentation). [Project page | PDF]

- Proposed a conditional WSGM for SBI; replaced DDPM with EDM (Karras) scheduling and added per-scale-normalized wavelet factorization.
- Achieved ∼73% faster sampling while cutting GPU memory use by ∼50% at matched fidelity.

B. Kurtkaya, **E. Çırakman**, *et al. Dynamical phases of short-term memory mechanisms in RNNs*. ICML 2025. [arXiv | ICML Entry]

- Revealed two STM mechanisms (slow-point manifolds vs. limit cycles) producing stable neural sequences without periodic drive.
- Derived power-law scaling of the critical learning rate with delay and mapped phase diagrams over learning rate, delay, and task design.

F. Dinc*, **E. Çırakman***, M. J. Schnitzer, H. Tanaka. *A ghost mechanism: An analytical model of abrupt learning*. Physical Review X (Accepted). [arXiv]

- Identified a ghost-induced transient bottleneck near a saddle-node remnant driving abrupt learning and timed working-memory delays.
- Derived the exact critical learning-rate law $\alpha^* \propto T^{-5}$ and a no-learning zone with oscillatory minima; validated trends across low- and full-rank RNNs.

C. Korkmaz, **E. Çırakman**, A. M. Tekalp, Z. Doğan. *Trustworthy SR: Resolving ambiguity in image super-resolution via diffusion models and human feedback*. ICIP 2024. [arXiv | ICIP Entry]

- Introduced human-in-the-loop selection and ensembling for diffusion SR to form ambiguity-aware, trustworthy reconstructions.
- Demonstrated that standard metrics (PSNR/SSIM/LPIPS/DISTS) can misalign with human trust under SR ambiguity, advocating preference-informed evaluation.

P. Ünal, **E. Çırakman**, *et al.* *A Big Data Application in Manufacturing Industry: Computer Vision to Detect Defects on Bearings*. IEEE Big Data 2022. [DOI]

- Built and deployed TC-VISION: a real-time CNN-based optical inspection system with a hardware and big-data pipeline for rolling-bearing quality control.

*Preprints & In Preparation*

**E. Çırakman**, *et al.* (In preparation for CVPR 2026). *Wavelet-Domain Image Prior Learning for Identity-Preserving Super-Resolution.*

- Developed a diffusion-based SR framework (SR3/EDM backbone) operating in the wavelet domain (SWT/WPT) to learn inter-band statistics of real data.
- Employed per-band KL divergence and FWD for distribution matching in high-frequency bands, ensuring preservation of text structure (OCR consistency) and face identity (ArcFace similarity) while suppressing artifacts.

**E. Çırakman**, *et al.* (In preparation). *Optimal Whitening Procedures for High-Frequency Detail Capture in Curvelet-Based Generative Models.*

- Investigated optimal whitening procedures within the curvelet transform framework to enhance high-frequency detail capture in generative models.

**E. Çırakman**, *et al.* (In preparation). *Heavy-Tailed Diffusion for Minority-Mode Coverage.*

- Explored heavier-tailed noise schedules to improve rare-structure coverage and address class imbalance, with initial results on well-log generation and rare geological features.

**E. Çırakman**, *et al.* (In preparation). *Curvelet-Adapted Diffusion and Curvelet Neural Operators.*

- Exploited curvelet sparsity and directional anisotropy for 2D/3D inversion pathways, incorporating 1D→2D guidance.

## Research & Industry Experience

Georgia Institute of Technology, SLIM Lab — *Dec 2024–Present*
*Research Intern* (Advisor: Prof. F. J. Herrmann)

- Built a cascaded multi-scale wavelet-domain posterior surrogate and ported DDPM to EDM (Karras) for improved stability and speed; generalized WSGM to the conditional case by factorizing $p(\mathbf{x} \mid \mathbf{y})$ across scales and deriving per-scale objectives with reverse-SDE samplers for better-conditioned scores.
- Developed tail-robust diffusion schedules for minority/outlier coverage and analyzed schedule–attention coupling in 1D well-log synthesis with 1D→2D guidance.
- Prototyped wavelet/curvelet-adapted diffusion and a Curvelet Neural Operator for geologically consistent samples.

Stanford University, CNC Program — *Jan 2024–Dec 2024*

*Research Intern* (Advisors: Prof. M. J. Schnitzer, Dr. F. Dinc)

- Formulated FORCE learning in a dynamical-systems framework; derived stability/chaos and memory-parameter scaling for RNNs, validated on timed working-memory tasks.
- Co-developed the ghost-mechanism theory; mapped slow-point vs. limit-cycle phases with large-scale diagnostics, identifying critical learning-rate scaling and no-learning zones.
- Built reproducible pipelines demonstrating that increasing trainable rank and reducing output confidence mitigate instabilities across low/full-rank RNNs.

Koç University, KUIS AI Center — *2021–Jan 2024*

*Research Intern* (Advisor: Prof. A. M. Tekalp)

- Customized SwinIR for single-image SR (attention/windowing, train/inference recipes, ablations) to improve a ViT baseline and benchmark capacity vs. fidelity trade-offs.
- Implemented diffusion SR with human-in-the-loop selection and ensemble decoding; designed evaluation protocols using universal IQA alongside traditional metrics.

TEKNOPAR, Ankara, Turkey — *2022–2023*

*Researcher*

- Built a real-time, patch-wise SVDD pipeline for anomaly detection in manufacturing processes.
- Implemented digital twins by integrating sensor telemetry with data-driven models for production monitoring.

## Awards & Achievements

- McClelland Scholarship, Stanford University (2024).
- Ranked in the top 0.7% nationally in Turkey's university entrance exam (YKS).
- SAUVC World Champion (2022) and RAMI Runner-up (2023) with ITU AUV Team (Vision & ROS).
- Teknofest AI Finalist (AI in transportation category).

## Selected Projects

*Cryptocurrency Price Forecasting (BERT+LSTM)*: Combined contextual embeddings with sequence models for short/long-horizon forecasts under high volatility.
*ITU AUV Team (Vision Pipeline)*: Built the vision pipeline and ROS integration for autonomous navigation.

## Skills

- **Programming**: Python (PyTorch, TensorFlow, OpenCV), C/C++, MATLAB.
- **Tools**: CUDA, Docker, Linux, ROS, Simulink, Devito, JUDI.

## Selected Coursework

Advanced: *Intro to Optimal Control*, *Discrete-Time Control & Formal Methods*, *Theoretical Machine Learning for EE*, *Control of Nonlinear Dynamic Systems.*